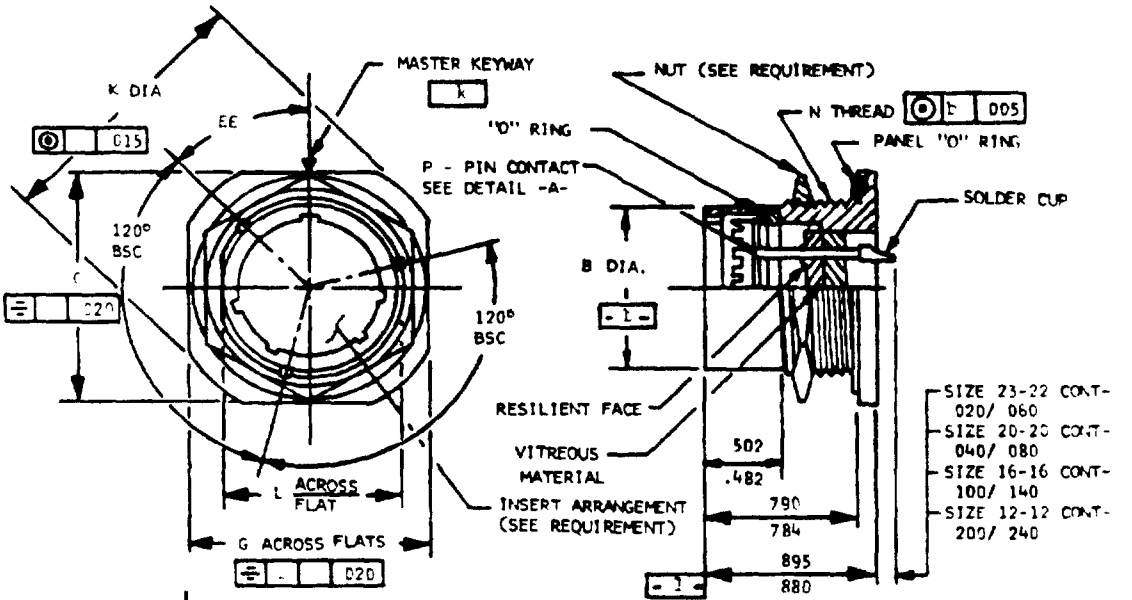
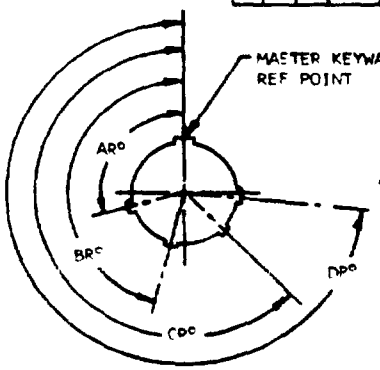
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT ENVIRONMENT RESISTING RECEPTACLE, JAM NUT MOUNT SOLDER-TYPE CONTACTS PINS ONLY CLASS D (SERIES 4).

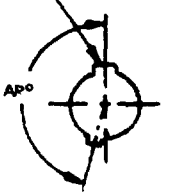
THE COMPLETE REQUIREMENTS FOR PROCURING THE CONNECTOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE OF SPECIFICATION MIL-C-0081511

THIS SPECIFICATION IS MANDATORY FOR USE BY ALL ACTIVITIES OF THE DEPARTMENT OF THE NAVY

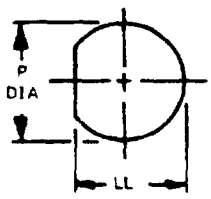




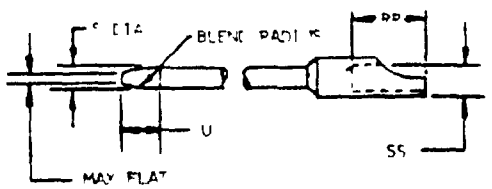
KEYWAY POSITION
NORMAL(1)POSITION
SHELL SIZE 10 AND LARGER
(SEE NOTE 5)



KEYWAY POSITION
NORMAL(1)POSITION
SHELL SIZE 8
(SEE NOTE 5)



PANEL CUTOUT
.125 MAX PANEL THICKNESS
(REF)



DETAIL -A-

INCHES	MM	INCHES	MM
.015	.38	.240	6.09
.020	.51	.515	13.02
.040	1.02	.635	16.13
.060	1.52	.917	23.29
.080	2.03	.923	23.44
.100	2.54	1.013	25.73
.125	3.18	1.028	26.11
.140	3.56		

FSC 5935

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TABLE I

SHELL SIZE	B DIA	G ACROSS FLAT	K DIA	L ACROSS FLAT	LL	N THREAD	P DIA
8	.553 (14.05) .547 (13.89)	.989 (25.12) .969 (24.61)	1.072 (27.23) 1.052 (26.72)	.598 (15.19) .590 (14.99)	.610 (15.49) .600 (15.24)	5/8-20UN-2A	.640 (16.26) .630 (16.00)
10	.678 (17.22) .672 (17.07)	1.114 (28.30) 1.094 (27.79)	1.197 (30.40) 1.177 (29.90)	.723 (18.36) .715 (18.16)	.735 (18.67) .725 (18.42)	3/4-20UNEF-2A	.765 (19.43) .755 (19.18)
14	.928 (23.57) .922 (23.42)	1.401 (35.59) 1.381 (35.08)	1.510 (38.35) 1.490 (37.85)	.973 (24.71 .965 (24.51)	985 (25.02) .975 (24.77)	1-20UNEF-2A	1.015 (25.78) 1.005 (25.53)
16	1.054 (26.77) 1.048 (26.62)	1.526 (38.76) 1.506 (38.25)	1.635 (41.53) 1.615 (41.02)	1.098 (27.89) 1.090 (27.69)	1.110 (28.19) 1.100 (27.94)	1 1/8-20UN-2A	1.140 (28.96) 1.130 (28.70)
18	1.178 (29.92) 1.172 (29.77)	1.651 (41.94) 1.631 (41.43)	1.745 (44.32) 1.725 (43.82)	1.222 (31.04) 1.214 (30.84)	1.235 (31.37) 1.225 (31.12)	1 1/4-20UN-2A	1.265 (32.13) 1.255 (31.88)

SHELL SIZE	AR° BSC	BR° BSC	CR° BSC	DR° BSC	PANEL 'O' RING REPLACEMENT	KEY POSITION (EE°BSC)(DEGREES) 1	2	3	4	5	6
8	128°				MS9068-018	40	40	40	40	40	40
10	105°	140°	215°	265°	MS9068-020	60	46	46	60	89	60
14	105°	140°	215°	265°	MS9068-024	60	46	46	60	89	60
16	105°	140°	215°	265°	MS9068-026	50	50	50	50	50	50
18	105°	140°	215°	265°	MS9068-028	50	50	50	50	50	50

CONTACT SIZE	S DIA	MAX FLAT	U	RR	SS
23-22	0275 (70) 0260 (66)	010 (25)	022 (56) 015 (38)	135 (3 43) 115 (2 92)	038 (97) 035 (89)
20-20	0410 (1 04) .0390 (.99)	.016 (41)	026 (66) 018 (46)	135 (3 43) 115 (2 92)	052 (1 32) 044 (1 12)
16-16	0630 (1 60) .0610 (1.55)	025 (.64)	026 (66) 018 (46)	.185 (4 70) .165 (4 19)	077 (1 96) 069 (1 75)
12-12	0950 (2 41) 0930 (2 36)	.062 (1 57)	026 (66) 018 (46)	185 (4 70) 165 (4 19)	125 (3 18) 114 (2 90)

NOTES
1. DIMENSIONS ARE IN INCHES
2. MILLIMETERS ARE IN PARENTHESES
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM
4. DIMENSIONS ARE AFTER PLATING
5. NORMAL MATING KEYWAY POSITION SHOWN. FOR ALTERNATE KEYWAY POSITION SEE MIL-C-0081511, FIGURE 4
6. DISTANCE FROM FRONT OF SHELL TO THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS PLUG SHELL, ENGAGES THE SPRING MEMBER
7. FOR SHELL SIZES 20, 22, AND 24 USE MIL-C-0081511/44 CONNECTORS

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REQUIREMENTS

DIMENSIONS AND CONFIGURATION SEE FIGURE AND TABLE I

SERIES 4

CLASSES D

SHELL SIZE SEE TABLE II

INSERT ARRANGEMENT SEE SUPPLEMENT FOR MS SHEET CORRESPONDING TO DESIRED SHELL SIZE.

CONTACTS STYLE P, SOLDER TYPE, SUP TERMINATION

Ⓐ CONTACT MATERIAL (SIZES 20 AND 23 ONLY): SIZE 20 AND 23 PIN CONTACTS SHALL BE MADE FROM HIGH STRENGTH NON-FERROUS ALLOY.

WEIGHT SEE TABLE II

Ⓐ CONTACT BEND STRENGTH: PIN CONTACTS SHALL WITHSTAND THE FORCES SPECIFIED IN TABLE III, APPLIED PERPENDICULAR TO THE CONTACT AXIS, AND SHALL NOT DISPLACE BEYOND THE LIMITS SPECIFIED.

SHELL POLARIZATION: 1 THRU 6 KEY POSITION

WEIGHT: SEE TABLE II

MARKING PART NUMBER ON SHELL

MATING COUNTERPART: PLUG CONNECTORS SPECIFIED IN MIL-C-81511/6, /34, AND MIL-C-0081511/56.

SLOTTED NUT: SHALL CONFORM TO MIL-C-81511/20.

PART NUMBER THE PART NUMBER SHALL CONSIST OF M81511/54A TOGETHER WITH THE APPLICABLE CODES FOR CLASS, SHELL SIZE, INSERT ARRANGEMENT, CONTACT STYLE, AND KEY POSITIONS. INFORMATION CONCERNING THE FORMULATION OF THE PART NUMBER CAN BE FOUND IN MIL-C-0081511.

TABLE II

SHELL SIZE	WEIGHT (REF) (POUNDS)
18 (CODE F)	.125
16 (CODE E)	.111
14 (CODE D)	.091
10 (CODE B)	.054
8 (CODE A)	.042

TABLE III

Ⓐ

CONTACT SIZE	MAX. DISPLACEMENT, IN.	FORCE, LBS.
23	.032	6
20	.032	10
16	.062	15
12	.062	20

CUSTODIAN
NAVY-AS

PREPARING ACTIVITY:
NAVY-AS

REVIEW ACTIVITY
NAVY-EC

(PROJECT NO. 5935-N929-6)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No 22-R293

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-M-52769 MEMBRANE, SURFACING: AIRFIELD, HEAVY DUTY

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO (If "yes" in what way?)

4 REMARKS (Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity Optional) | DATE

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